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Development and Assessment of Dual-Fuel Capabilities in Next-Generation Aero-Engine Injectors for Hydrogen and Liquid Fuels

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Dual-fuel (DF) aero-engines, capable of using hydrogen alongside sustainable aviation fuel (SAF) or kerosene as a fuel, would be a key enabler in the transition to hydrogen operation but have received limited attention in research and development. Therefore, this study explores the potential of retrofitting the kerosene injector in a Rolls-Royce Pearl Rich Quench Lean (RQL) combustor for dual-fuel operation. Adapting the existing injector for the combined use with hydrogen poses a significant challenge due to elevated NO_x emissions. To address this, an automated computational fluid dynamics (CFD) workflow is used to optimize the injector geometry for emission performance, enabling rapid evaluation under realistic engine conditions. To explore a broad range of geometry variations, the Reynolds-Averaged Navier–Stokes (RANS) approach is employed for simulations, while large eddy simulations (LES) are performed on selected geometries to validate emission trends. Prior to optimizing the injector, the CFD setup is validated against available measurements for conceptual injector geometries, confirming its capability to accurately predict emission levels. This study demonstrates that dual-fuel capability can be incorporated into an existing combustion system while preserving flame stability and emission levels comparable to standard kerosene injectors. The CFD-based workflow successfully identified optimal geometric features, achieving a reduction of emissions for both fuels during Landing-Take-Off cycle for the proposed injector configuration. Emissions are evaluated in accordance with the International Civil Aviation Organization (ICAO)-prescribed legislation. In the future, the optimized injector will undergo further evaluation through experimental testing to confirm the predicted performance and set the next step toward dual-fuel capable combustion systems. [DOI: 10.1115/1.4069797]

Keywords: hydrogen, injection, combustion, dual fuel, emission, optimization

1 Introduction

Hydrogen has gained significant attention as a sustainable fuel for reducing carbon dioxide emissions from aero engines [1,2]. However, the widespread adoption of hydrogen faces significant challenges, including limited availability and infrastructure gaps in many regions. Dual-fuel (DF) capable engines provide a decisive solution for the transition to hydrogen, allowing the seamless integration of hydrogen without sacrificing operational flexibility. Due to hydrogen's lower volumetric energy density and the added weight of its fuel tanks, long-haul flights are expected to continue relying on liquid fuels such as kerosene and SAFs. In this context, DF engines provide the flexibility to take off with hydrogen and switch to liquid fuels midflight, effectively eliminating soot and CO_2 emissions during airport residence.

Research on dual-fuel combustion systems is limited, primarily focusing on reciprocating engines [3], with an emphasis on pollutant emissions and the ignition mechanisms of natural gas–air mixtures. Their potential for gas turbine or aero-engine applications has received little attention [4]. A fundamental study by Vance et al. [5] demonstrates that blending hydrogen with jet fuel improves flame stability and offers the potential to reduce carbon dioxide emissions. However, as an initial step, this study focuses on the single fuel cases of operating with either pure hydrogen or pure kerosene, excluding fuel blend scenarios.

While hydrogen combustion has the potential to reduce carbon dioxide and particulate emissions, nitrogen oxides (NO_x) emissions are a primary concern due to the higher adiabatic flame temperatures and distinct properties of hydrogen as a fuel.

At equivalent power outputs, systems using conventional fuels can generate significantly lower NO_x emissions compared to those

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burning hydrogen-containing fuels, particularly when rich-quench-lean combustors are utilized. For instance, a numerical study [6] demonstrated that blending hydrogen into natural gas during burner operation results in large increases in NO_x emissions. Increasing the hydrogen content in the fuel mixture can lead to a rise in NO_x emissions by up to a factor of four. These findings underscore the importance of carefully controlling NO_x emissions in hydrogen-based combustion systems.

Reducing emissions in dual-fuel combustors is particularly challenging due to the need to balance tradeoffs between various pollutants, in particular NO_x and soot. For instance, rich combustion with shorter residence times lowers NO_x emissions by reducing peak temperatures but increases soot emissions due to incomplete combustion. In contrast, longer residence times promote NO_x formation but allow for more complete combustion and reoxidation, thereby reducing particulate emissions like soot. Hence, optimizing the injector's geometry to create targeted flow patterns and flame structures within the combustor provides an effective method for reducing overall emissions.

Building on these considerations, current efforts focus on modifying the Pearl 15 engine, designed around the Rich Quench Lean (RQL) concept, to incorporate an injection system compatible with both hydrogen and liquid fuels. This retrofit approach seeks to minimize costs and expedite development. This study aims to advance these developments by adapting a fully automated computational fluid dynamics (CFD) framework to evaluate and optimize fuel injectors for dual operation with hydrogen and kerosene/sustainable aviation fuel, while minimizing emissions.

The remainder of this work is structured as follows: First, the baseline design for the dual-fuel injector is presented. This is followed by a description of the automated CFD framework and the methodology for optimizing the NO_x and soot emissions. It includes the automated meshing process, CFD setup, and a data-driven optimization approach. Following validation of the framework using available experimental data, the optimization results are presented and analyzed. Finally, the conclusions and outlook are given.

2 Baseline Design for the Dual-Fuel Injector

Before optimization, a base design of the DF injector, capable of operating with both fuels, is introduced. Based on this initial configuration, parameters are identified that can be optimized under specific constraints to minimize emissions. A schematic of the base injector is provided in Fig. 1. As shown by the arrows, the design features central injection for sustainable aviation fuel/kerosene and hydrogen injection positioned slightly further outward.

The simulation of the baseline dual-fuel injector operating with kerosene, shown in Fig. 2, provides insight into how the resulting flame stabilizes within the combustor. It presents contour plots of temperature and velocity profiles under full-load conditions. Two distinct combustion zones are observed: a rich zone located upstream of the dilution ports and a lean zone formed downstream of the secondary air injection.

The velocity and temperature structures observed are highly similar between the standard kerosene injector and the dual-fuel injector operating with kerosene, as the geometry modifications to the kerosene injector are minimal. Achieving a similar flow field and flame topology is essential for hydrogen operation.

As this work aims to optimize the design for minimal emissions with both fuels, the following outlines a set of geometric and operational constraints, which are also applied to the baseline design, defining the range within which the optimizer can operate.

2.1 Dimensions of the Injector. The design space for the DF injector is defined by the constraints imposed by the existing Pearl combustor geometry. The available volume determines the maximal geometrical dimensions of the injector, while the required power output at takeoff dictates the required hydrogen mass flow. To ensure consistent turbine inlet temperatures for both hydrogen and kerosene, the air-to-fuel ratio is adjusted, resulting in an air-to-fuel ratio approximately three times higher for hydrogen compared to kerosene. To minimize the impact on kerosene operation, the kerosene passage, and prefilmer retain their standard geometry, while the inner air passage and swirler are resized to allow accommodation of the hydrogen passage.

2.2 Flame Liftoff. To prevent flame stabilization on the injector tip, maintaining a high axial flow velocity is critical. High shear rates at the injector exit promote local quenching, facilitating a lifted flame. An effective strategy to achieve sufficient inlet velocity is to reduce the swirl in the outer air channel. This minimizes the tangential velocity components and pressure drop, directly contributing to higher axial flow velocities. However, reducing the swirl necessitates further design adaptations, as swirl plays a crucial role in stabilizing the primary combustion zone. It enhances fuel-air mixing and generates centrifugal forces, which create a low-pressure region along the centerline. This low-pressure region allows for the recirculation of hot combustion gases, which are essential for maintaining flame stability. In configurations with reduced swirl, stability is ensured through the design of outward-directed air and fuel channels, which create a diverging flow pattern essential for backflow and hot gas recirculation. Therefore, the opening angle must be carefully selected to maintain an open flame for both hydrogen and kerosene operation under reduced swirl conditions.

2.3 Effective Area. Finally, to maintain the airflow distribution of the combustion system, it is critical to closely match the effective areas of the dual-fuel injector with those of the standard injector. The effective area is defined by the air mass flow through the injector and the pressure drop across it.

$$A_{\text{eff}} = \frac{\dot{m}}{\sqrt{2\rho\Delta p}} \tag{1}$$

2.4 Geometric Parameters for the Optimization. Building on the previously discussed geometric and operational constraints, a preliminary sensitivity analysis identified three geometric parameters as key factors significantly influencing flow and combustion behavior within the combustion chamber (CC). As illustrated in Fig. 1, the key parameters include **Channel Height** h of the outer air channel, which determines the air distribution between the primary and secondary combustion zones; **Opening Angle** α of the outer air channel, which impacts the size of the recirculation zone; and **Vane Angle** β of the outer swirler, which affects the mixing behavior within the chamber.

The optimization of the baseline geometry to minimize emissions is restricted to acceptable parameter levels that follow the previously outlined constraints. The considered levels, normalized by their

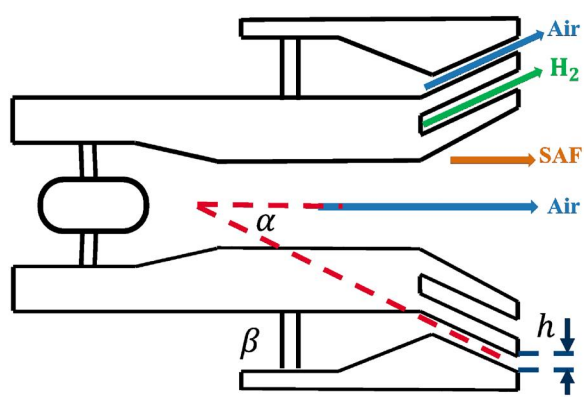

Fig. 1 Schematic of the dual-fuel injector configuration. Air passages are indicated by blue arrows, hydrogen passages by green arrows, and kerosene passages by red arrows. (Color version online.)

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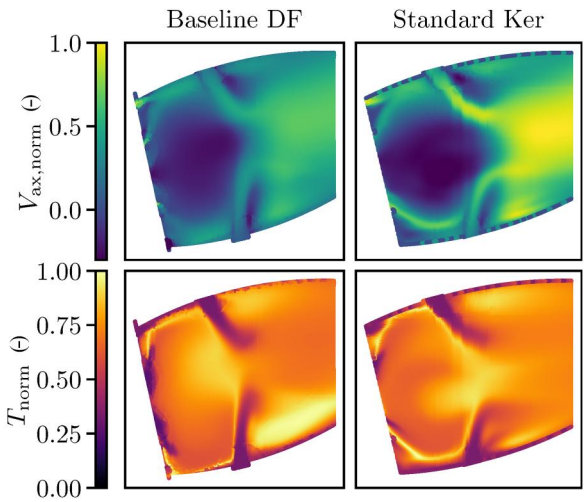

Fig. 2 Velocity and temperature contours for the baseline DF injector during kerosene operation and for the standard kerosene injector at full-load conditions. The contours represent a planar slice through the center of the domain.

respective maximum simulated values, are shown in Table 1. The values corresponding to the baseline dual fuel injector geometry before optimization are highlighted.

Bold values indicate the baseline geometry.

To investigate potential cross-correlational effects between the optimization parameters, a full factorial design of parameter variations is chosen for the study. This design includes all possible combinations of levels across all factors [7], resulting in 100 distinct parameter combinations.

Investigating a large number of geometries requires a structured and automated optimization approach for the injector design. A manual process for mesh generation across numerous geometries is inefficient and time-intensive, while manual data management challenges productivity and systematization. To address this, an automated workflow is developed to manage mesh generation, CFD simulations, data postprocessing, and optimization with minimal human involvement. The details of this setup are provided in the methodology section.

3 Computational Fluid Dynamics-Based Optimization Methodology

The adapted workflow integrates several components: a parameterized CAD geometry defined in Siemens NX, automated mesh generation with ICEM CFD software, CFD simulations with PRECISE-UNS solver, and data-driven postprocessing of the CFD results. Figure 3 provides an overview of the workflow, with each stage detailed in the sections below.

3.1 Geometry Parameterization and Automated Meshing. The parameterization of the CAD model for the injector and the CC involves defining their design through a set of rules and logical relationships. Instead of setting fixed dimensions, key components of the injector and CC are adjustable with constraints applied for the whole assembly using mathematical equations. These equations govern the relationships between geometric parameters, ensuring design flexibility while maintaining functional consistency. The automated geometry modification is achieved by using Visual Basic Scripting in combination with the expressions functionality of Siemens NX. An interface is established with the meshing software to enable streamlined and automated mesh generation. The meshing process employs the Octree algorithm [8], which initially generates a tetrahedral mesh for the entire computational domain. Subsequently, tetrahedral cells are selectively merged into hexahedral cells, producing a hex-dominant mesh. This method proves to be both computationally efficient and robust, significantly reducing the need for user intervention while maintaining a good mesh quality throughout the process. The resulting meshes comprise approximately 25×10^6 cells, with refinement applied in high-shear regions to accurately capture critical flow features, as determined through mesh independence studies. The mesh refinement regions are defined using the mesh density feature in ICEM CFD, with positioning of the refined mesh regions dynamically linked to the applied geometrical modifications. The required level of refinement was determined through testing progressively finer meshes, with results being considered mesh-independent when changes in key output emission metrics were negligibly small (less than 1%).

Table 1 Levels of normalized design parameters

Opening angle α_{norm}	Air channel height h_{norm}	Swirler angle β_{norm}
0.64, 0.73, 0.82, **0.90**, 1	0.17, **0.33**, 0.53, 0.67, 1	**0**, 0.6, 1

Bold values indicate the baseline geometry.

3.2 Computational Fluid Dynamics Simulations and Emissions Data Collection. The simulations in this study are conducted using the PRECISE-UNS CFD solver developed by Rolls-Royce [9]. To minimize computational cost, the Reynolds-Averaged Navier–Stokes (RANS) approach is employed, where all turbulent structures are modeled, reducing the required grid resolution. The k–ϵ realizable model is used for turbulence modeling [10].

The combustion process is modeled using a tabulated chemistry approach, which employs precomputed one-dimensional flamelets stored in a lookup table to determine thermo-chemical states during the simulation [11]. For kerosene, the progress variable is defined as a linear combination of the mass fractions of CO_2, H_2, and H_2O, whereas for hydrogen, it is represented by the mass fraction of H_2O. Flamelet calculations for the table generation are performed using CHEM1D [12] with a mixture-averaged transport model and an n-Dodecane reaction mechanism. The same reaction mechanism is used for both hydrogen and kerosene to facilitate a direct NO_x comparison. A comparison done in Ref. [5] shows that there is a good agreement between the hydrogen-specific mechanisms and the n-Dodecane mechanism for the predicted laminar burning velocities.

A presumed probability density function approach is used for turbulence-chemistry interaction modeling [13]. The subgrid-scale distribution is modeled using a beta distribution for the mixture fraction and a tri-delta distribution for the progress variable [14]. This results in a four-dimensional chemistry lookup table, comprising the mixture fraction, progress variable, and their respective variances.

The CFD simulations are conducted for a single sector of the combustion chamber using boundary conditions that reflect realistic operating conditions for the dual-fuel system. The main focus of this study is Idle and Take-off like conditions. To model the single sector, rotational cyclic boundary conditions are applied at the sector interfaces. The turbulent kinetic energy at the inlet is initialized based on values obtained from a compressor simulation, ensuring a realistic turbulence level in the flow. The evolution of the fuel spray for kerosene is modeled using a Lagrangian particle tracking approach [15]. Droplets are introduced at specified injection points, with their sizes determined based on a Rosin–Rammler distribution characterized by defined statistical parameters. A steady-state evaporation model is employed to simulate the droplet behavior. The resulting spray pattern is assumed to represent the distribution following the atomization process [14]. The statistical spray characteristics in the simulations are derived from those of the standard kerosene injector, as the prefilmer geometry remains unchanged in the dual-fuel configuration.

NO_x emission are calculated solving an additional transport equation, with source terms extracted from the chemistry table [16]. The soot model used in this study is the modified two-equation model based on the framework developed by Wen et al. [17]. This

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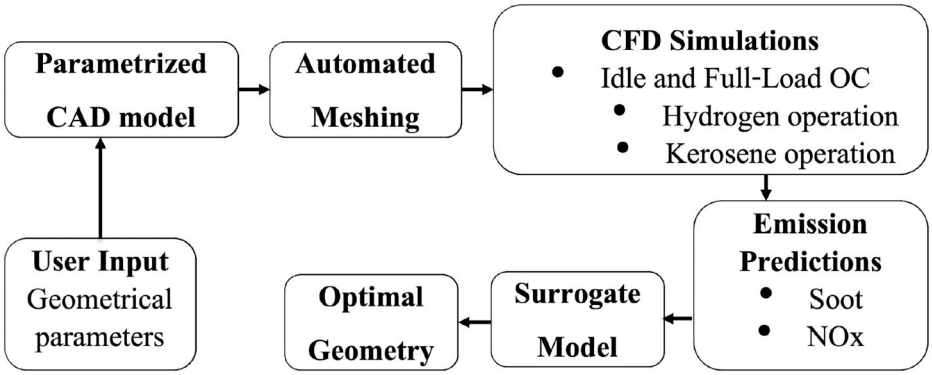

Fig. 3 Automated optimization workflow

approach involves solving two transport equations: one for the soot mass fraction and another for the soot number density.

Emission metrics in form of integrated NO_x and soot massflows at the combustor exit, are gathered from each simulation run. This emissions data is stored along with corresponding injector geometry parameters, establishing a structured dataset for the optimization process.

The pressure drop and air mass flow through the injector, necessary for calculating the effective area, are determined via CFD simulations. Consequently, certain preselected parameter levels are excluded from the optimization space if the simulations reveal that the effective area requirement is not met.

3.3 Surrogate Modeling With Gaussian Process Regression. In the subsequent stage, a fit using a Gaussian Process Regression algorithm [18] is performed to the optimization dataset obtained from the CFD simulations to create a continuous model enabling the estimation of emissions for any new configuration within the defined parameter bounds.

A Gaussian Process is a probabilistic model used to describe an unknown function $f(\mathbf{x})$ by assuming that any finite set of function values follows a multivariate Gaussian distribution. The model is defined as

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \tag{2}$$

where $m(\mathbf{x})$ is the mean function and $k(\mathbf{x}, \mathbf{x}')$ is the kernel function. The kernel is a combination of a radial basis function and a constant kernel:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\| \mathbf{x} - \mathbf{x}' \|^2}{2\ell^2}\right) + \sigma_n^2 \tag{3}$$

where σ_f^2 is the data variance, ℓ is the length scale, and σ_n^2 is the noise variance. The length scale ℓ represents the characteristic distance in the input space over which the function values (NO_x or soot massflows) notably change. This distance captures how sensitive the output (emission mass flows) is to variations in the input features (geometrical parameters). This approach allows fitting the data points without requiring a predefined fitting function.

A separate Gaussian Process Regression model is trained for each operational condition, fuel type, and emission type on corresponding subset of the collected data, resulting in six submodels. The submodel outputs are integrated into a full surrogate model using an objective function, which summarizes emission levels across fuel types and operational conditions, allowing for overall emission minimization.

3.4 Objective Function and Optimization. The objective function is based on the emission legislation for the Landing and Takeoff cycle prescribed by International Civil Aviation Organization (ICAO) [19]. According to the legislation, operating modes are weighted by corresponding time coefficients t_j: 26 min for idling and 42 s for takeoff. The equation for the objective function is further extended by incorporating the probabilities of fuel usage under a certain operating mode P_U, j and a weighting coefficient W_i, j for the emission massflow $\dot{M}_{i,j}$, allowing prioritization of optimization for soot or NO_x emissions.

$$\mathrm{E}(\mathrm{x}) = \sum_{i=1}^{2}\sum_{j=1}^{2} t_j \left(W_{i,j}\dot{M}_{i,j}(\mathbf{x})\right) P_{U,j} \tag{4}$$

Bayesian optimization is used to find the input parameters $\mathbf{x} = [\alpha, \beta, h]$ that minimize the total emissions $E(\mathbf{x})$.

The resulting integrated workflow is capable to identify optimal geometry with reduced NO_x and soot emissions across different usage scenarios. The schematics of the workflow is summarized in Fig. 3.

4 Results and Discussion

The results of the comprehensive CFD study are presented next. Prior to delving into the details of the optimization study, the CFD setup is validated against available experimental data.

4.1 Validation of the Computational Fluid Dynamics Setup. In a preliminary investigation to evaluate the feasibility of hydrogen operation, various hydrogen injector geometries with distinct topologies were tested in a full-annular rig featuring the Pearl700 combustor at the German Aerospace Center (DLR) HBK5 test facility in Cologne, Germany [20].

The tests encompassed full range of operating conditions, including idle, approach, climb, and takeoff.

Although these experimental data were collected using injectors different from those studied here, they provide a suitable dataset for validating the CFD setup, ensuring its reliability in accurately predicting emission trends based on injector geometry and operating conditions. The comparison between experimental and numerical NO_x predictions are presented in Fig. 4. Geometry 1 is similar to the hydrogen injector configuration used in the present study, but without the centrally integrated kerosene injector. Geometry 2 features central hydrogen injection, while geometries 3 and 4 incorporate discrete jet-based hydrogen injection with different jet distribution. A more detailed description of the test rig, tested geometries, and measurement approach can be found in Ref. [20].

The CFD simulations show a consistent ranking of the geometries compared to the experimental measurements.

The discrepancy observed under full-load conditions—where CFD predicts Geometry 1 to underperform relative to Geometry 3, whereas measurements suggest the opposite—can be attributed to the small absolute difference between their NO_x levels. Given the measurement uncertainties, the ranking remains in good qualitative agreement with the CFD results, indicating that CFD effectively

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captures the impact of injector geometry and operating conditions on emission level trends.

To further evaluate the CFD setup, Large Eddy Simulations (LES) are conducted for the full-load hydrogen operating case, focusing on air channel height variations of the baseline dual-fuel injector geometry. To model unresolved subgrid scales in LES, the Sigma model is used [21], which relates the SGS stresses to the singular values of the resolved velocity gradient tensor. A 3D velocity profile, obtained from a compressor simulation, is used in LES as a boundary condition for the inlet air to introduce a realistic level of turbulence into the flow.

Figure 5 presents a comparison, showing that the LES simulation predicts higher absolute NO_x emission levels. This is attributed to the high temporal dynamics of the resolved scales, which reveal a more intense mixing field with fuel-rich pockets and high-temperature regions at stoichiometry, promoting NO_x formation. Although this behavior can theoretically be captured with an accurately predicted subgrid probability density function shape, the complex mixing and flow patterns pose a significant challenge for presumed probability density function models. Despite these challenges, the consistent trends observed between RANS and LES simulations confirm the RANS approach's capability to reliably predict overall emission trends.

For simulations of kerosene combustion, the computational setup used in the optimization process corresponds to the methodology outlined in Ref. [14]. A thorough validation of this CFD framework, including comparisons with LES and experimental results for a generic combustion chamber, is presented in Ref. [14].

4.2 Sensitivity Analysis Findings. Using the previously validated setup, the simulations of the parametric variations presented in Table 1 are performed. The results of the sensitivity analysis for single-parameter variations are presented in Fig. 6. For each geometrical group, only one parameter of the baseline geometry is varied at a time.

The emission indices for soot and NO_x are normalized by the corresponding EI value of the baseline geometry for each pollutant. The analysis reveals the tradeoffs among different emission types, fuels, and operating conditions. The distinct combustion characteristics of hydrogen and kerosene are reflected in opposing trends and differing slopes for the predicted NO_x curves. Kerosene operation reveals a clear tradeoff between reducing soot and minimizing NO_x emissions. Conflicting trends are also observed under different operating conditions, showing that designs optimized for a specific operating point are likely to deliver suboptimal performance under alternative conditions.

Hydrogen combustion emissions exhibit a higher sensitivity to geometric modifications, with potential emission reductions of up to 20% under idle conditions and 8% under full-load conditions. This is primarily attributed to the significantly elevated adiabatic flame temperatures associated with hydrogen combustion. Since NO_x formation is highly temperature-dependent, even small variations in the temperature field, given these elevated adiabatic temperatures, can lead to pronounced changes in NO_x emissions. Figure 7 presents normalized temperature profiles, averaged over the radial cross section of the combustion chamber, plotted against the normalized distance from the injection point for the variations of the opening angle. The temperature normalization is based on the maximum radial cross section averaged temperature observed during hydrogen operation. Figure 7 shows considerably higher temperatures within the primary combustion zone for hydrogen operation.

During kerosene operation, fluctuations in soot emissions are more pronounced than variations in NO_x mass flow. This is primarily due to the strong dependence of soot formation on the local gas-phase composition. Even minor changes in fuel-air mixing can lead to significant response in terms of soot production, as soot formation exhibits a nonlinear increase with richer fuel mixtures. Unlike NO_x, which is primarily governed by thermal conditions and residence time, soot formation is highly dependent on mixing dynamics and the presence of hydrocarbon-rich regions that promote the formation of soot precursors.

Among the parameters tested, the outer air channel swirl has the least impact on emissions. This limited influence can be attributed to the dominant effect of the swirler in the central air channel. In the following discussion, the results of the opening angle variation and air channel height variation are examined in more detail to analyze the underlying physics of the observed emission fluctuations. The effects of swirl are omitted for brevity.

4.3 Effects of the Variation in the Injection Angle α_{norm}. Figures 8(*a*) and 8(*b*) depict the temperature field's dependence on the opening angle for hydrogen and kerosene, respectively. The depicted temperature fields represent planar slices through the center of the domain. The white line represents the isoline of the stoichiometric equivalence ratio for each fuel. Analyzing the mixing behavior and resulting temperature profiles indicates that the observed trends are heavily influenced by the mixing processes within the combustion chamber.

Below the temperature contour plots in Figs. 8(*a*) and 8(*b*), bar charts illustrate the equivalence ratio distributions within the flame region, conditioned by elevated heat release rates, showing the proportions of lean, stoichiometric, and rich mixtures in the primary zone (PZ), secondary zone (SZ), and the entire combustor. Additionally, the averaged equivalence ratio are given for each zone, providing a quantitative assessment of the mixing and combustion behavior for both fuels. The dashed red line marks the stoichiometric equivalence ratio.

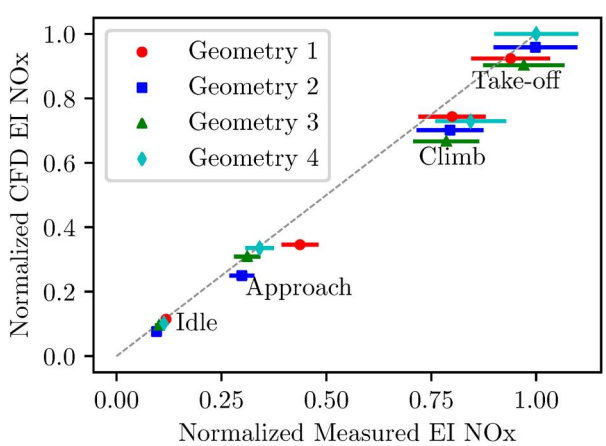

Fig. 4 Comparison of measured EI NO_x values with CFD-predicted EI NO_x for conceptual dual-fuel geometries across the full range of operating conditions

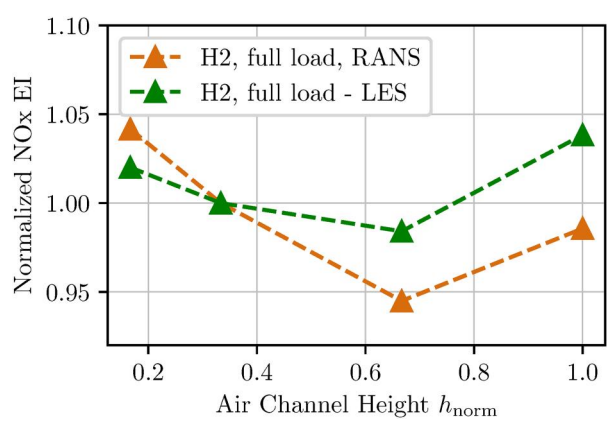

Fig. 5 Comparison of NO_x emission trends predicted by LES and RANS for a series of geometries with varying air channel heights during hydrogen operation. Normalization is performed using the corresponding values for the baseline geometry.

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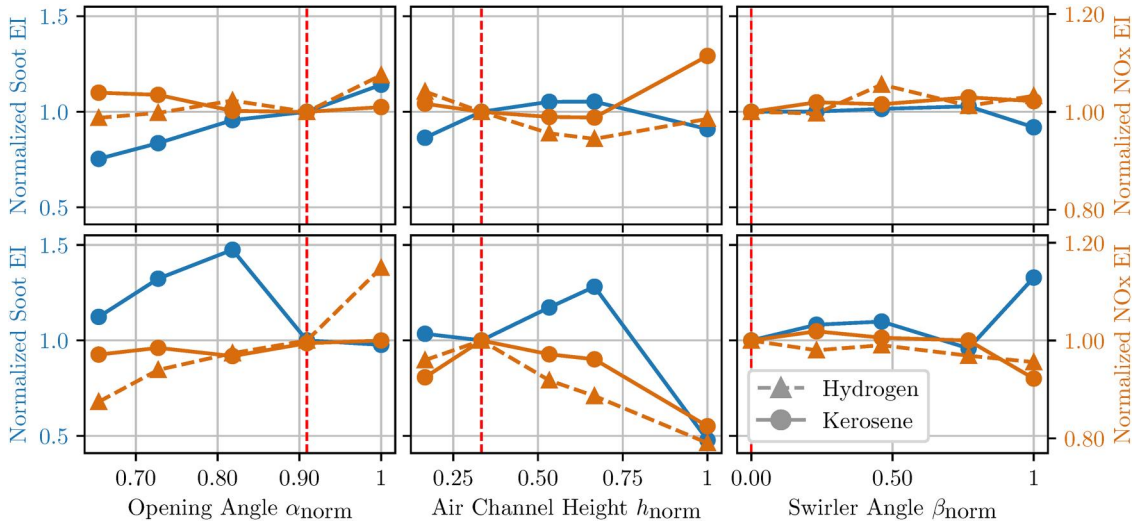

Fig. 6 Sensitivity of emissions to parametrical adaptations for full load (TOP) and idle (BOTTOM) operating conditions. NO_x EI shown in orange, soot EI in blue color. Normalization is performed using the corresponding values for the baseline geometry. (Color version online.)

As shown in Fig. 8(*a*) the distributions indicate that for hydrogen reaction predominantly occurs under lean conditions in both the primary and secondary zones. Overall, a bimodal behavior is observed, with peaks around equivalence ratios closer to stoichiometric $\phi \approx 0.86$ and at lean conditions around $\phi \approx 0.36$. The primary combustion zone (PZ), characterized by elevated temperatures, is the predominant region for NO formation. However, to fully account for the observed NO_x trends, contributions from both the primary and secondary combustion zones must be considered.

As the injection angle increases, the recirculation zone in the primary combustion region expands, and the proportion of near-stoichiometric equivalence ratios increases, leading to a rise in the mean temperature within this zone, as illustrated in Fig. 7. Simultaneously, combustion occurs over a larger volume, enhancing hydrogen consumption in the primary zone and resulting in a leaner secondary combustion zone. The overall trend for NO_x emissions shows an increase for normalized opening angles (α_{norm}) ranging from 0.64 to 0.82 due to the rising temperature in the primary zone. At $\alpha_{norm} = 0.9$, a local minimum in NO_x emissions is observed, as the higher production of NO_x in the primary zone is partially counterbalanced by the reduced emissions in the secondary zone.

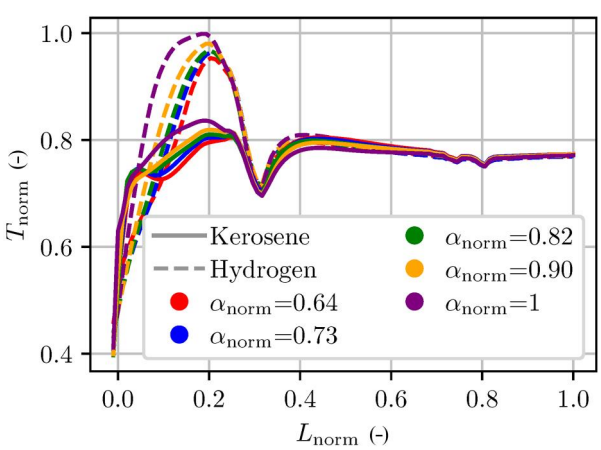

Fig. 7 Normalized temperature averaged across the radial cross section of the combustion chamber as a function of distance from the injection point for the opening angle variation. Normalization done with the maximum temperature during hydrogen operation.

At the maximum injection angle, the radial velocity component becomes dominant, resulting in the formation of two distinct flame branches along the walls and an increased proportion of near-stoichiometric conditions, as evident in the contour plot for the maximum value $\alpha_{norm} = 1$. Consequently, this shift in local equivalence ratios significantly amplifies overall NO_x emissions.

For kerosene operation, shown in Fig. 8(*b*), the combustion process follows the RQL principle. Increasing the injection angle enhances mixing and results in a reduction in the fraction of highly rich equivalence ratios. Specifically, the peak at an equivalence ratio of $\phi \approx 2.5$ disappears in the distribution, with the equivalence ratio shifting closer to stoichiometry from the rich side. As a result, the temperature in the primary zone increases, similar to hydrogen operation, a trend that is also clearly reflected in Fig. 7. Simultaneously, the secondary zone becomes leaner and cooler. Figure 7 illustrates that, for kerosene operation, unlike for hydrogen, temperature levels in the primary and secondary zones are comparable. This observation suggests that both zones contribute comparably to the overall NO_x production, particularly when considering the extended length of the combustor beyond the dilution ports, which amplifies the influence of the secondary zone. The overall trend in NO_x emissions for injection angle variation for kerosene shows a decrease, as demonstrated in Fig. 6. This is due to the pronounced decrease in NO_x formation in the secondary zone outweighing the increase observed in the primary zone.

Another crucial aspect to consider is soot formation. As illustrated in Fig. 9, the reduced momentum—and consequently, the lower shear and turbulence levels—significantly affect local mixing, as reflected by the increased variance in the mixture fraction. This heightened variance suggests a greater likelihood of forming fuel-rich pockets near the injector, which accounts for the substantial increase in soot production.

4.4 Effects of the Variation in the Air Channel Height h_{norm}. The variation in air channel height influences the air split between the primary and secondary airflows while maintaining a constant total airflow rate into the combustion chamber. Figures 10 and 11 present normalized temperature fields and NO_x source terms for different air channel heights, with temperature fields shown as planar slices through the central axis of the domain. The white line represents the isoline of the stoichiometric equivalence ratio for each fuel.

For the smallest air channel height, for kerosene operation, NO_x formation is highly localized. Defined regions of NO_x formation can be observed in the secondary zone, where, during the quenching

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process, the stoichiometric mixture fraction is reached due to insufficient mixing rates in the primary zone and a larger amount of unconsumed fuel being transported downstream.

As the air channel height increases, enhanced mixing occurs due to the larger introduced momentum and air mass, leading to a broader and more gradual reaction zone where combustion progresses at lower average temperatures. Although NO_x formation is distributed over a larger volume within both the primary and secondary combustion zones, an overall reduction in NO_x emissions is observed as shown in Fig. 6. This can be attributed to the slower NO_x formation kinetics by decreased temperature, which mitigates the impact of the extended zone of formation.

Nearly complete fuel consumption within the primary zone at the maximal value of the air channel height $h_{norm} = 1$ is effectively

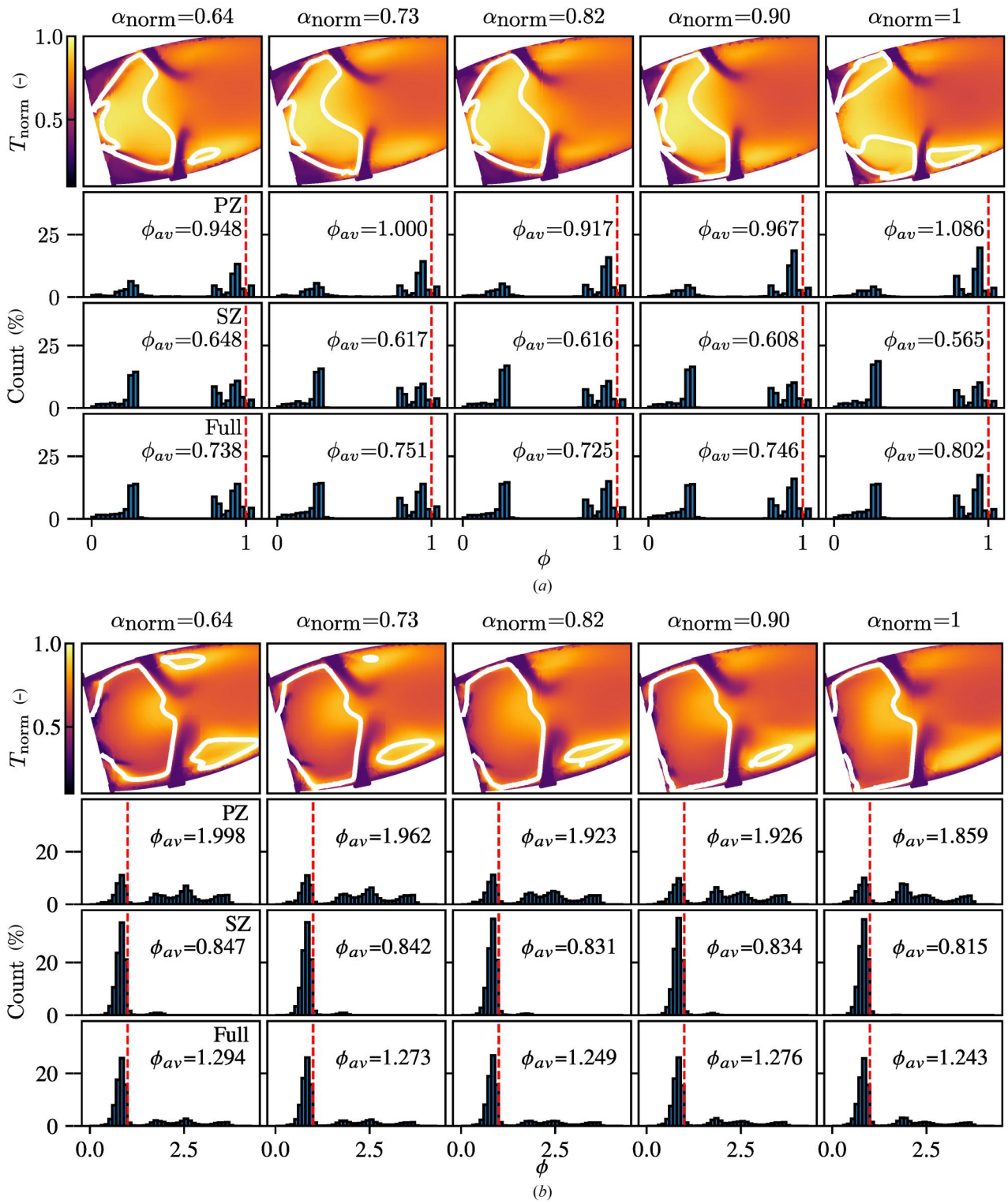

Fig. 8 Influence of the outflow angle α_{norm} on temperature field and equivalence ratio within flame regions conditioned by elevated heat release rates. Analyzes done under full-load operating condition in the primary zone (PZ), secondary zone (SZ), and the entire combustion chamber (full). (Color version online.)

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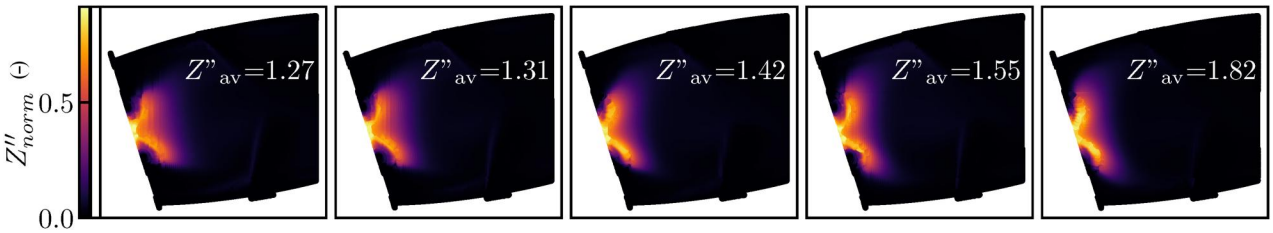

Fig. 9 Influence of the outflow angle α_{norm} on variance of the mixture fraction under full-load operating condition for kerosene. Normalized opening angle values form left to the right are 0.64, 0.73, 0.82, 0.9, 1.

suppressing NO reduction through hydrocarbon radicals in the secondary zone. In the numerical simulations, this is reflected in low NO_x destruction rates after the dilution ports. Consequently, for the largest air channel height ($h_{norm} = 1$), NO_x emissions increase significantly, as NO_x formed in the primary zone is no longer mitigated by subsequent reduction mechanisms.

Hydrogen combustion, shown in Fig. 11, is characterized by more distributed NO_x production due to its higher diffusivity, which promotes enhanced mixing. The effect of varying the air channel height closely resembles that observed with kerosene. As the airflow in the primary combustion zone increases, the mean temperature decreases in both the primary and secondary combustion zones. In the primary zone, this reduction is due to leaner fuel-air mixtures and intensified mixing caused by the increased momentum of the injected air. In the secondary zone, the temperature drop is a result of increased fuel consumption in the primary zone, leading to leaner mixtures downstream of the dilution ports, as well as improved homogenization of the mixture and temperature fields.

In summary, the implemented geometric modifications result in physically consistent changes in pollutant formation, validating the model's suitability and enabling the full factorial parameter variation to construct a surrogate model based on these data.

4.5 Full Factorial Parameter Variation and Response Surface. Figure 12 presents the results of the full factorial parameter variation performed under full-load operating conditions. The dataset for idle operating conditions is not shown for the sake of brevity. The analysis includes 60 geometries out of the initially considered 100 geometries that are found to meet the constraints on effective area and axial velocity outlined in the section on the baseline geometry. The design space is constrained to a maximum normalized outer air channel height of 0.67, as higher values exceed the allowable injector effective area. The swirler vane angle is restricted to a maximum scaled value of 0.76 to prevent excessive tangential velocity components. Emission index values have been normalized by the highest value observed for a given emission type and operational load across all geometries. Consistent with the data representation in Fig. 6, the kerosene results are shown as dots, while hydrogen results are represented by triangles.

This analysis illustrates interaction effects of combined geometrical modification and provides a comprehensive overview of the design space. Distinct trends are evident for each geometrical parameter variation, with the air channel height—which influencing the air mass injected into the primary zone—emerging as a particularly significant factor. While key trends from the sensitivity analysis are generally replicated in the full factorial numerical experiment, certain specific effects are not captured. This suggests that optimization based solely on One Factor at a Time tests may lead to the selection of a local minimum rather than an overall optimal geometry. Notably, increasing the air mass introduced into the primary zone (normalized air channel height values between 0.33 and 0.53) consistently reduced emissions for both fuels. Furthermore, adjustments to swirl levels reduced the impact of other parameters, as demonstrated by smaller emission level variations in high-swirl cases. The resulting dataset from the full factorial study both for full load and idle operating conditions is utilized for model training. The complete surrogate model, developed using the obtained data and the previously described approach, is presented in Fig. 13. It represents a function that links the combined emission levels, calculated in accordance with ICAO legislation, to the geometric adaptations. Equal weights are assigned to the optimization for hydrogen and soot emissions, with equal probabilities of usage considered for both fuels under both operating conditions.

For validation of the surrogate predictions, a test is performed on a geometry that is not involved in the fitting process. The results, shown in Fig. 14, compare the EI predicted by the fitted model to the CFD output, along with the uncertainty intervals. The predicted soot emission index shows slightly higher values for both operating conditions but remains in good agreement with the CFD results.

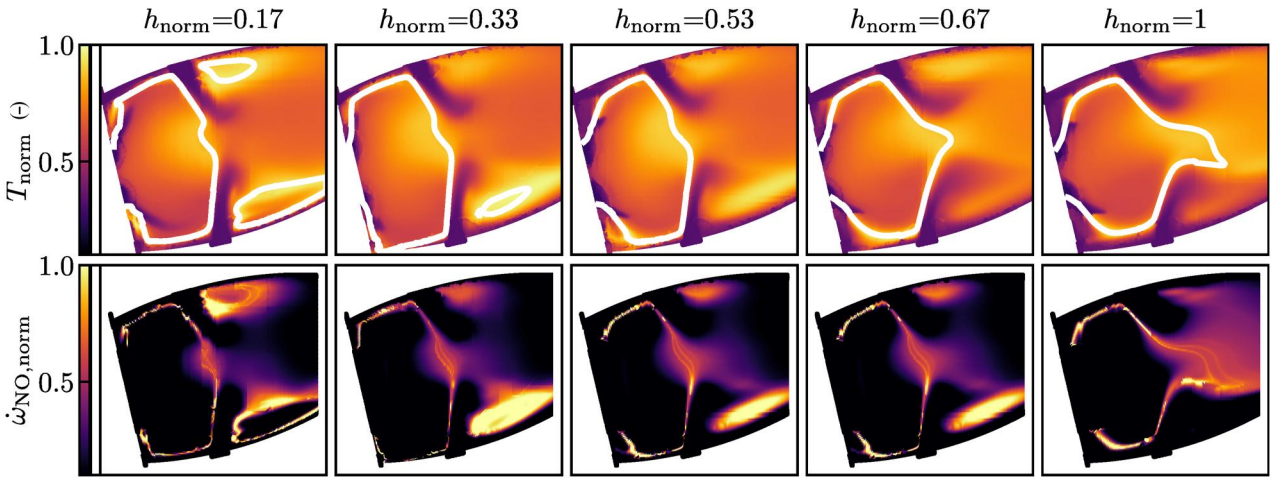

Fig. 10 Influence of the air channel height h_{norm} on temperature field and NO_x formation zones under full-load operating condition for kerosene

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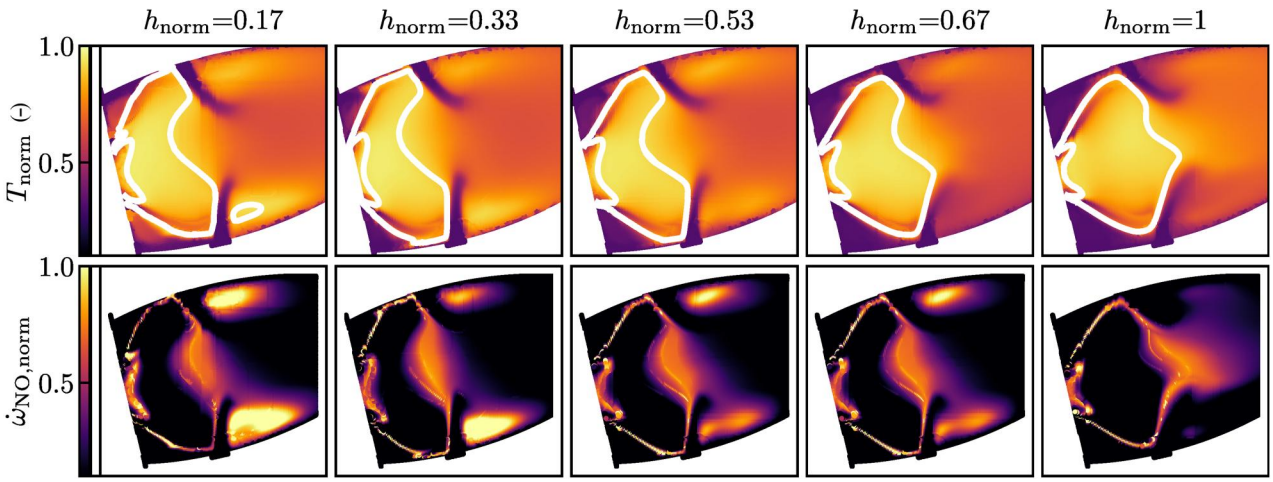

Fig. 11 Influence of the air channel height h_{norm} on temperature field and NO_x formation zones under full-load operating condition for hydrogen

The optimization results recommend selecting the largest outer air channel variant within the design space, a shallow injection angle, and low swirl for the outer air channel.

Table 2 provides an overview of emission levels predicted by CFD and, where available, obtained from measurements. All emissions are expressed as percentages relative to the standard kerosene injector's values. The comparison includes four injector configurations: the standard kerosene injector, the baseline DF injector, the optimized DF injector, and Geometry 1—a hydrogen-only injector with an identical topology to the hydrogen injection system of the DF injector. Geometry 1 corresponds to the dual-fuel injector with the kerosene channel omitted and is characterized by normalized parameters: $\alpha_{norm} = 0.64$, $\beta_{norm} = 0$, and $h_{norm} = 0.2$.

The optimized dual-fuel injector achieves reduced emission mass flows across all considered emission types and operational conditions compared to the baseline design, with the sole exception of NO_x emissions during kerosene operation at full load. This exception results from an unavoidable tradeoff between conflicting optimization objectives.

Experimental measurements for Geometry 1 indicate that hydrogen operation results in NO_x emissions that are 20% higher under full-load conditions and 46% higher under idle conditions compared to kerosene operation with the standard kerosene injector at the same combustor exit temperature. However, given that the combined NO_x emissions for the standard kerosene injector remain well below the maximum ICAO-prescribed limits—and considering the complete

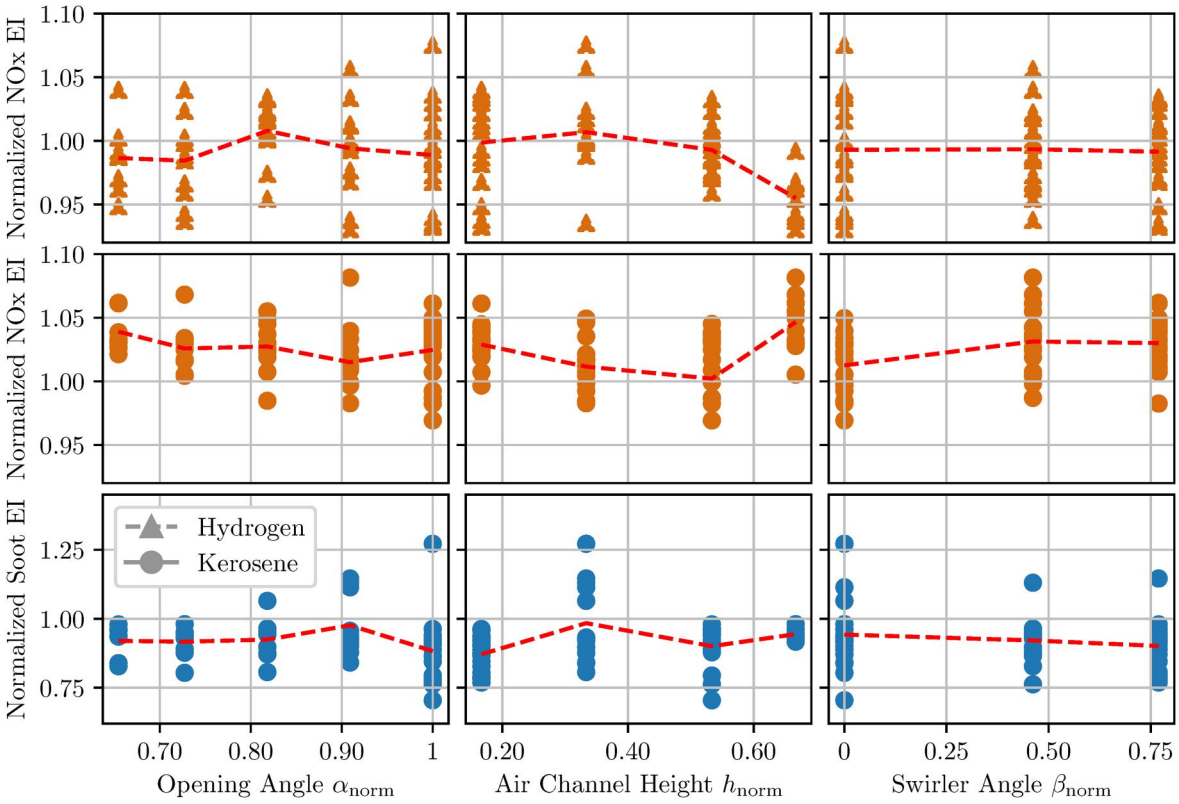

Fig. 12 Results of a full factorial parameter variation under full-load operating conditions. Mean trend shown with dashed red line. Normalization is performed using the corresponding values for the baseline geometry. (Color version online.)

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elimination of CO and soot emissions during hydrogen operation—Geometry 1, even before optimization, remained within regulatory compliance.

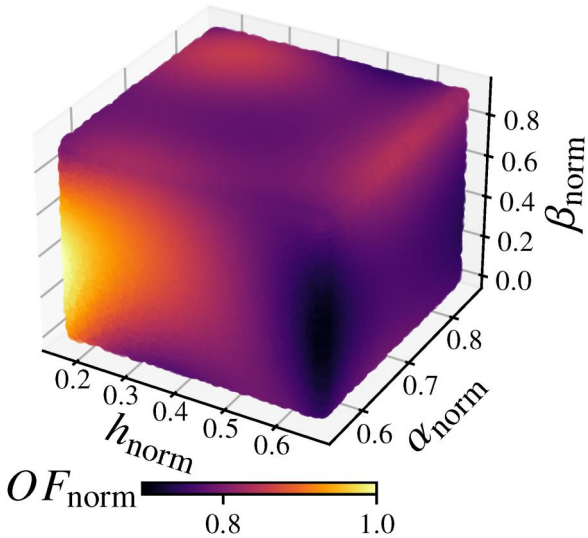

Fig. 13 Visualization of the objective function with equal probabilities of kerosene and hydrogen usage, and equal weighting assigned to soot and NO_x optimization

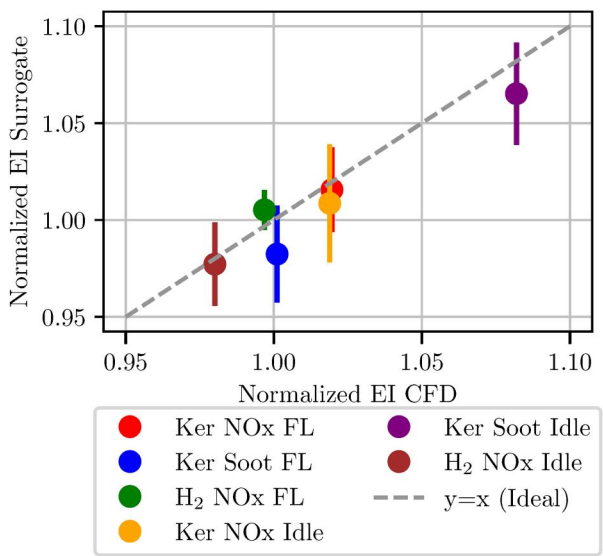

Fig. 14 Comparison of the surrogate Emission Index (EI) predictions for a test geometry ($\alpha_{norm} = 0.9$, $\beta_{norm} = 0.3$, $h = 0.33$). Normalization with respect to the corresponding EI values for the baseline geometry.

Following the optimization of the dual-fuel injector NO_x emissions were reduced to levels even lower than those of Geometry 1 for hydrogen operation, as demonstrated by CFD results. Additionally, soot emissions during kerosene operation were further reduced, achieving levels lower than those of the standard kerosene injector.

These results clearly showcase the potential of designing a dual-fuel injector that, after geometric optimization, can match or even surpass the performance of established single-fuel injectors. This is an exciting development, as it paves the way for retrofitting existing architectures to support dual-fuel operation, enabling a faster transition to low- and zero-CO_2 emission aviation.

5 Conclusion

The pressing demand for fuel infrastructure-independent aero engines, facilitating a swift transition to a hydrogen-based aviation economy, motivated the evaluation of dual-fuel capable injectors in this study.

An automated optimization workflow for a DF injector has been developed, integrating a parameterized CAD model with a RANS CFD setup coupled to tabulated chemistry and state-of-the-art emission models. The CFD framework has been successfully validated against experimental and LES data, demonstrating a general overprediction of NO_x emissions, while maintaining consistency with experimental trends. This workflow has enabled efficient injector optimization with a focus on minimizing emissions.

An evaluation of 60 geometries has indicated that all tested designs comply with current legislative limits. The analysis has revealed a significant sensitivity of NO_x emissions to injector geometry during hydrogen operation, with an optimization potential of up to 20%. Emissions are strongly influenced by the mixing behavior within the combustion chamber. While hydrogen operation has resulted in higher absolute NO_x emissions compared to kerosene, these increases can be partially mitigated through geometric optimization.

This work demonstrates that retrofitting a dual-fuel injector into an existing combustion chamber, combined with a systematic optimization approach, can effectively achieve NO_x emission standards mandated by legislation for both hydrogen and kerosene.

A comprehensive investigation of the spray behavior will be conducted as part of future work to validate the assumption that it remains identical to that of a standard kerosene injector. This analysis will involve high-fidelity numerical simulations using the Volume of Fluid method to evaluate key parameters such as droplet size distribution, velocity profiles, and atomization dynamics.

Additionally, future work will focus on manufacturing and testing of the proposed injector, as well as further examining the emission characteristics during blended-fuel operation, as well as during fuel switching scenarios.

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Table 2 Comparison of CFD and measured (Mes) emission values for baseline DF geometry, optimized DF geometry, and concept Geometry 1 under full load (FL) and idle conditions to standard kerosene injector

Injector (α_{norm}, β_{norm}, h)	OP	CFD Soot: Ker	CFD NO_x: Ker	CFD NO_x: H_2	Mes NO_x: Ker	Mes NO_x: H_2
Standard Ker	FL (%)	100	100	—	52.9	—
(-, -, -)	Idle (%)	100	100	—	66.9	—
Baseline DF	FL (%)	142	106	87.2	—	—
(0.9, 0, 0.33)	Idle (%)	96.3	119	155	—	—
Optimized DF	FL (%)	97.8	114	83.9	—	—
(0.64, 0.4, 0.67)	Idle (%)	80.1	116	131	—	—
Geometry 1	FL (%)	—	—	88.8	—	63.4
(0.64, 0, 0.2)	Idle (%)	—	—	134	—	98

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Funding Data

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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